

Trial Higher School Certificate Examination

2002



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1-10
- All questions are of equal value

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 (12 marks)

- a) Evaluate $\frac{23.97 - (3.62)^2}{\sqrt{4.51}}$ correct to 2 decimal places 1
- b) Solve: $\frac{x}{3} - \frac{2x+1}{4} = 5$ 2
- c) Sketch $y = \sin 2x$ for $0 \leq x \leq 2\pi$ 2
- d) Solve: $|1 - 2x| < 4$ 3
- e) Evaluate $\log_e 3.5$, correct to 2 decimal places 1
- f) If $\frac{dy}{dx} = x^2 + 8x$ and when $x = 3$, $y = 0$ find y in terms of x 2

Question 2 (12 marks)

a) Differentiate:

(i) $y = (2x - 5)^4$ 2

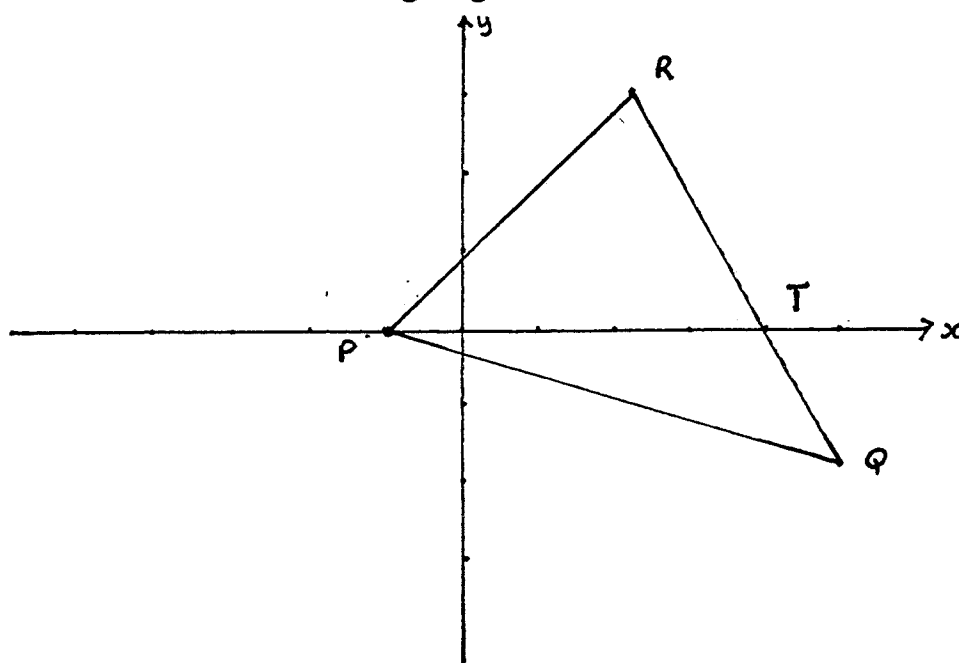
(ii) $y = e^{2x} \sin x$ 2

b) Using the information on the following diagram

$P(-1, 0)$

$R(2, 3)$

$Q(5, -2)$



(i) Show that the equation of the line PQ is $x + 3y + 1 = 0$ 2

(ii) Find the length of PQ 1

(iii) Find the perpendicular distance from R to PQ 2

(iv) Find the area of $\triangle PRQ$ 1

(v) Find the size of the angle RTP correct to nearest degree 2

Question 3 (12 marks)

a) Prove $\operatorname{cosec} x \cos x \tan x = 1$

1

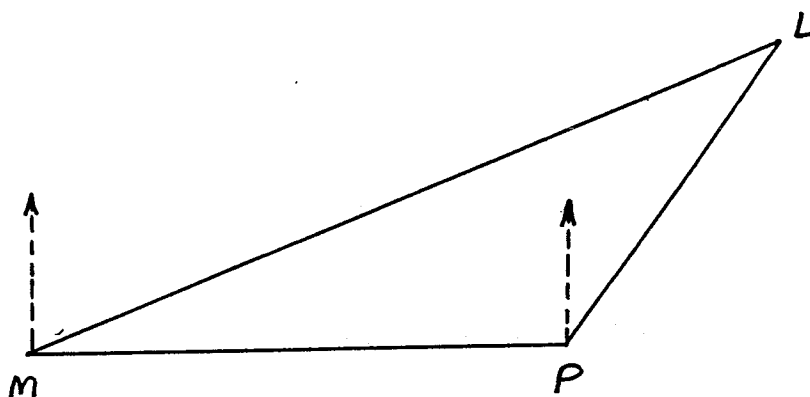
b) Given that $\cot \beta = \frac{3}{2}$ and $\sin \beta < 0$ find exact value of $\cos \beta$

2

c) Solve $2\sin^2 x + \cos x = 2$ for $0^\circ \leq x^\circ \leq 360^\circ$

3

d) The bearing of a lighthouse L from a ship at M is $N55^\circ E$. The ship then sails due East from M to a point P which is 10 nautical miles from L . The bearing of the lighthouse from P is $N25^\circ E$.



(i) Copy the diagram into your answer booklet.

(ii) Deduce that $\angle MLP = 30^\circ$

1

(iii) Show that $MP = 5 \operatorname{cosec} 35^\circ$ and hence find MP correct to 1 decimal place.

3

(iv) If the ship continues to sail due East from P find its shortest distance from the lighthouse.

2

Question 4 (12 marks)

a) Find the primitive of:

(i) $3xe^{x^2}$ 2

(ii) $\tan x$ 2

b) (i) Sketch the curve $y = \ln(4 - x)$ showing the x and y intercepts 2

(ii) Find the equation of the tangent at the point where the curve crosses the x axis 2

(iii) Find the exact area bounded by the curve and the x and y axes 4

Question 5 (12 marks)

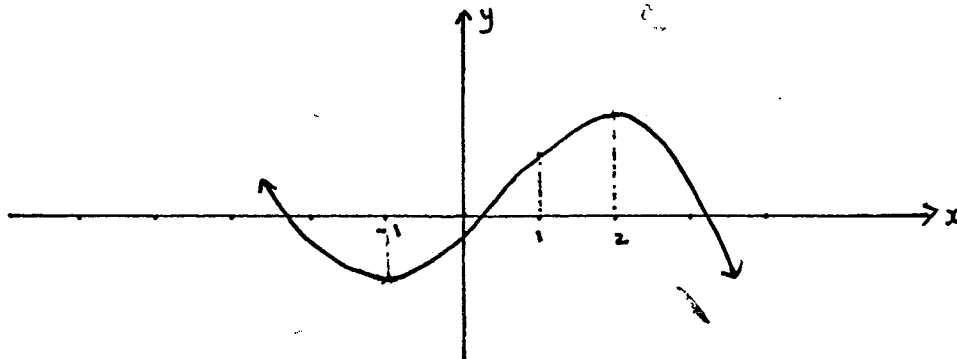
- a) Using the following table of values find an approximation to the value of:

2

$\int_1^5 f(x) dx$ using Simpson's Rule with 5 function values

x	1	2	3	4	5
$f(x)$	1.74	3.9	4.2	7.89	10.2

- b) The following is a graph of some function $y = f(x)$.



- (i) In your answer booklet graph $y = f'(x)$ and $y = f''(x)$ if at $x=1$ there is a point of inflexion.

3

- (ii) Explain what happens to the curve at a point of inflexion.

1

- c) (i) Determine the stationary points and their nature for the curve

$y = x^3 - 6x^2 + 9x - 7$

4

- (ii) Sketch the curve

1

- (iii) For what x values is the curve increasing?

1

Question 6 (12 marks)

- a) Zinc is extracted from a mine at a rate that is proportional to the amount of zinc remaining in the mine. Hence the amount R remaining after t years is given by $R = R_0 e^{-kt}$ where k is a constant and R_0 is the initial amount of zinc. After 5 years, 50% of the initial amount of zinc remains.

(i) Find the value of k (correct to 4 decimal places)

2

(ii) How many more years will elapse before only 30% of the original amount remains?

3

- b) Joanne decided to save for an overseas holiday. She decided to deposit \$500 into a special account at the beginning of each month for 3 years. The account paid 6% pa compounded monthly.

(i) How much is the first payment of \$500 worth at the end of 3 years?

1

(ii) Prove, by developing a geometric series, that the total value of all her deposits at the end of 3 years is given by

$$\text{Total value} = 100\,500 (1.005^{36} - 1)$$

3

(iii) Calculate this value.

1

(iv) If Joanne had needed a lump sum of \$22 000 by the end of the third year what would she have had to save each month? (You may use any previous working).

2

Question 7 (12 marks)

- a) The limiting sum of the infinite geometric series $1 + 5^x + 5^{2x} + \dots$ is 5. Find x (correct to 3 decimal places). 3
- b) Find the x values of the intersection points of the curves $x - y = 4$ and $y = x^2 - 3x - 4$. Hence, find the area between the two curves. 4
- c) (i) Write down the discriminant of: $3x^2 + 2x + k$. 2
- (ii) For what values of k does $3x^2 + 2x + k = 0$ have real roots?
- d) If one root of the equation $mx^2 - px + 1 = 0$ is double the other, prove that $2p^2 = 9m$. 3

Question 8 (12 marks)

- a) The velocity of a particle moving in a straight line is given by $v = 2t - 6$ where position is measured in metres and time in seconds. If the particle is initially 4 metres to the right of the origin
- (i) Find an expression for displacement. 1
 - (ii) Find when and where the particle comes to rest. 2
 - (iii) Find the total distance covered by the particle in the first 5 seconds. 1
 - (iv) Evaluate $\int_0^5 (2t - 6) dt$ 2
 - (v) Explain why your answers for parts (iii) and (iv) are not the same. 1
- b) The rate at which gas escapes from a balloon is given by $\frac{dG}{dt} = \frac{-3}{t+1}$ where the gas is measured in cm^3 and time in seconds.
- (i) Find G as a function of time if the initial amount of gas in the balloon is 10cm^3 . 2
 - (ii) How long before all the gas has escaped? 3

Question 9 (12 marks)

a) Find:

(i) $\int \sin^2 x \cos x \, dx$

1

(ii) $\int_0^{\frac{\pi}{4}} (2 \sin x - \cos 2x) \, dx$

3

b) Sketch the curve $y = 2 \cos \pi x$ for $0 \leq x \leq 2$

1

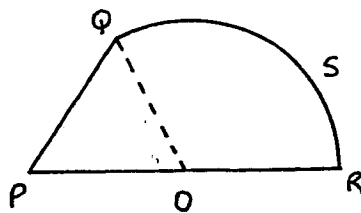
On the same set of axes sketch the graph $y = 1 - x$

1

Using your graph state how many solutions $2 \cos \pi x = 1 - x$ has in the domain $0 \leq x \leq 2$

1

c) The region $QPORS$ is formed by an equilateral triangle OPQ with a side of 12cm and a sector $QORS$. PR is a straight line. QSR is an arc of the circle centre O .



Giving answers in exact form find:

(i) The perimeter of the region

2

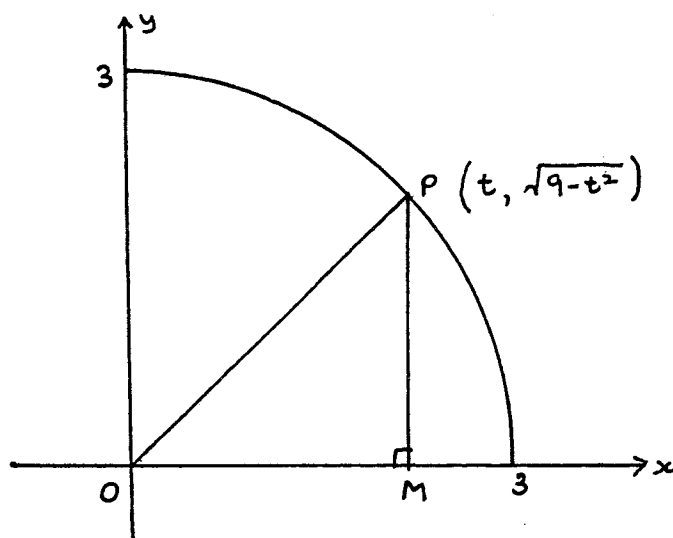
(ii) The area of the region

3

Question 10 (12 marks)

- a) (i) State domain and range of the function $y = \sqrt{9-x}$ 2
- (ii) Sketch a graph of this function 1
- (iii) Calculate the volume of the solid generated when the area bounded by the curve and the coordinate axes in the first quadrant is rotated about the y axis. 3

- b) The diagram shows the curve $y = \sqrt{9-x^2}$ for $x \geq 0$. P is the point $(t, \sqrt{9-t^2})$ on the graph and M is the foot of the perpendicular from P to the x axis.



- (i) Write down an expression in terms of t for the area A of the triangle OPM . 1
- (ii) Find the coordinates of the point P which gives triangle OPM a maximum area. 5

QUESTION 1:

(a) 5.12 (correct to 2 dec. pl)

(b) $\frac{x}{3} - \frac{2x+1}{4} = 5$

$$\frac{4x - 3(2x+1)}{12} = 5$$

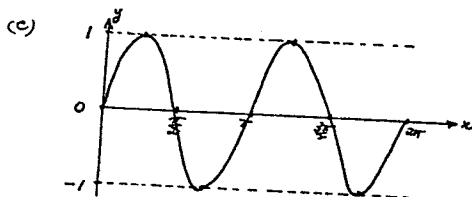
$$\frac{4x - 6x - 3}{12} = 5$$

$$\frac{-2x-3}{12} = 5$$

$$-2x-3 = 60$$

$$-2x = 63$$

$$\therefore x = -\frac{63}{2}$$



(d) $|1-2x| < 4$

$$\Rightarrow -4 < 1-2x < 4$$

$$-5 < -2x < 3$$

$$\frac{5}{2} > x > -\frac{3}{2}$$

$$\therefore -\frac{3}{2} < x < \frac{5}{2}$$

2

(e) $\log_e 3.5 = 1.25$ (correct to 2 dec. pl)

(f) $\frac{dy}{dx} = x^2 + 8x$

$$\Rightarrow y = \frac{x^3}{3} + 4x^2 + C$$

when $x=3, y=0$

$$\therefore 0 = 9 + 36 + C$$

$$\therefore C = -45$$

$$\therefore y = \frac{x^3}{3} + 4x^2 - 45$$

3

QUESTION 2:

(a) (i) $y = (2x-5)^4$

$$\frac{dy}{dx} = 4(2x-5)^3 \cdot 2$$

$$= 8(2x-5)^3$$

(ii) $y = \frac{e^{2x} \sin x}{x}$

$$\frac{dy}{dx} = v u' + u v'$$

$$= \sin x \cdot 2e^{2x} + e^{2x} \cos x$$

$$= e^{2x}(2\sin x + \cos x)$$

(b) (i) $P(-1, 0)$ $Q(5, -2)$

$$m_{PQ} = \frac{-2-0}{5-(-1)}$$

$$= -\frac{2}{6}$$

$$= -\frac{1}{3}$$

 $\therefore$ Equation of PQ is

$$y - 0 = -\frac{1}{3}(x + 1)$$

$$3y = -x - 1$$

$$\therefore x + 3y + 1 = 0$$

(ii) $PQ = \sqrt{(5-(-1))^2 + (-2-0)^2}$

$$= \sqrt{36 + 4}$$

$$= \sqrt{40}$$

$$= 2\sqrt{10}$$

4

(iii) $(2, 3)$ to $x + 3y + 1 = 0$

$$d = \frac{|1(2) + 3(3) + 1|}{\sqrt{1^2 + 3^2}}$$

$$= \frac{12}{\sqrt{10}}$$

$$= \frac{12\sqrt{10}}{10}$$

$$= \frac{6\sqrt{10}}{5}$$

$$(iv) \text{ Area of } \triangle PRQ = \frac{1}{2} \times PQ \times d$$

$$= \frac{1}{2} \times 2\sqrt{10} \times \frac{6\sqrt{10}}{5} \text{ (units}^2\text{)}$$

$$= 12 \text{ (units}^2\text{)}$$

(v) $m_{QR} = \frac{3-2}{2-5}$

$$= -\frac{1}{3}$$

$$\therefore \hat{RTP} = \tan^{-1}\left(\frac{1}{3}\right)$$

$$= 59^\circ \text{ (correct to nearest minute)}$$

QUESTION 3:

$$(a) \operatorname{cosec} x \cos x \tan x = \frac{1}{\sin x} \times \frac{\cos x}{1} \times \frac{\sin x}{\cos x} = 1$$

$$(b) \left. \begin{aligned} \cot \beta &= \frac{3}{4} \\ \sin \beta < 0 \end{aligned} \right\} \Rightarrow \beta \text{ is in 3rd quadrant}$$

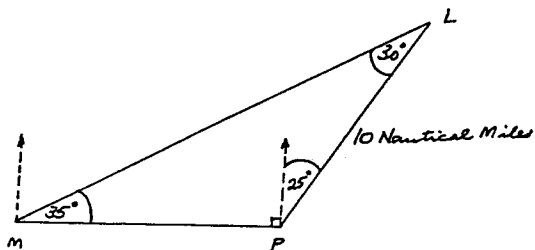
$$\therefore \cos \beta = -\frac{3}{5}$$

$$\begin{aligned} (c) \quad 2 \sin^2 x + \cos x &= 2 & 0^\circ \leq x \leq 360^\circ \\ 2(1 - \cos^2 x) + \cos x &= 2 \\ 2 - 2 \cos^2 x + \cos x &= 2 \\ 2 \cos^2 x - \cos x &= 0 \\ \cos x (2 \cos x - 1) &= 0 \\ \therefore \cos x &= 0, \frac{1}{2} \\ \cos x = 0 &\Rightarrow x = 90^\circ, 270^\circ \\ \cos x = \frac{1}{2} &\Rightarrow x = 60^\circ, 300^\circ \end{aligned}$$

$$\therefore x = 60^\circ, 90^\circ, 270^\circ, 300^\circ$$

6.

(d)



$$(i) \angle MNP = 180^\circ - (35^\circ + 90^\circ + 25^\circ) = 30^\circ \quad (\text{angle sum of } \Delta \text{ is } 180^\circ)$$

$$(ii) \text{ In } \Delta MNP \text{ by the Sine Rule}$$

$$\frac{MP}{\sin 30^\circ} = \frac{10}{\sin 35^\circ}$$

$$\therefore MP = \frac{10 \sin 20^\circ}{\sin 35^\circ}$$

$$= 5 \operatorname{cosec} 35^\circ$$

$$= 8.717...$$

$$= 8.7$$

(correct to 1 dec. place)

$$(iii) \text{ Using Cosine Rule}$$

$$LM^2 = (8.717)^2 + 10^2 - 2 \times 8.717 \times 10 \cos 115^\circ$$

$$= 249.66536$$

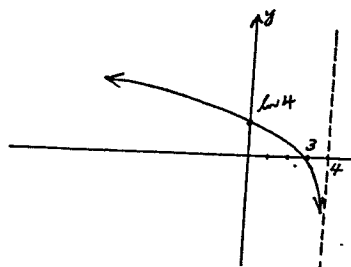
$$\therefore LM = 15.80 \text{ miles (to 2 d.p.)}$$

QUESTION 4:

$$(a) (i) \int 3x e^{x^2} dx = \frac{3}{2} \int 2x e^{x^2} dx = \frac{3}{2} e^{x^2} + C$$

$$\begin{aligned} (ii) \int \tan x dx &= \int \frac{\sin x}{\cos x} dx \\ &= - \int \frac{-\sin x}{\cos x} dx \\ &= -\log(\cos x) + C \end{aligned}$$

$$(b) (i) y = \ln(4-x) \quad \text{Domain: } 4-x > 0 \Rightarrow x < 4$$



$$\begin{aligned} x \text{ intercept at } y &= 0 \\ \text{ie } \ln(4-x) &= 0 \\ 4-x &= 1 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} y \text{ intercept at } x &= 0 \\ \text{ie } y &= \ln 4 \end{aligned}$$

8.

$$(ii) y = \ln(4-x)$$

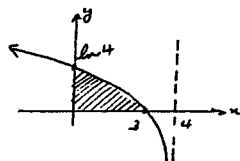
$$\Rightarrow \frac{dy}{dx} = \frac{-1}{4-x}$$

$$\begin{aligned} \text{at } (3, 0) \quad \frac{dy}{dx} &= \frac{-1}{4-3} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \therefore \text{Tangent at } (3, 0) \text{ is} \\ y - 0 &= -1(x - 3) \\ y &= -x + 3 \\ \text{or } x + y - 3 &= 0 \end{aligned}$$

$$(iii) A = \int_0^3 \ln(4-x) dx$$

but as we are unable to find the primitive of $\ln(4-x)$ we must refer the area to the y -axis as shown.



$$A = \int_0^{\ln 4} x dy$$

$$\begin{aligned} y &= \ln(4-x) \\ \Rightarrow e^y &= 4-x \\ \text{ie } x &= 4 - e^y \end{aligned}$$

$$= \int_0^{\ln 4} (4 - e^y) dy$$

$$= [4y - e^y]_0^{\ln 4}$$

$$= (4 \ln 4 - e^{\ln 4}) - (0 - 1)$$

$$= 4 \ln 4 - 4 + 1$$

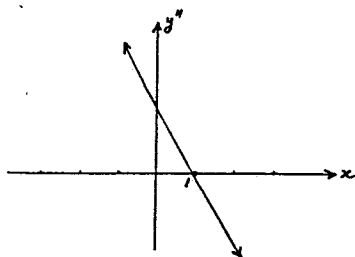
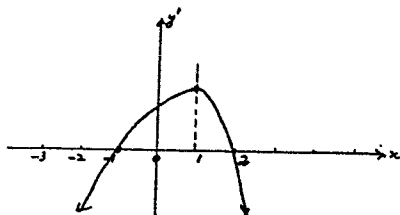
$$= 4 \ln 4 - 3$$

$$\text{Area is } (4 \ln 4 - 3) \text{ units}^2$$

QUESTION 5:

$$\begin{aligned}
 (a) \quad \int_1^5 f(x) dx &\doteq \frac{1}{3} [(2f_1 + f_5) + 4(f_2 + f_4) + 2f_3] \\
 &= \frac{1}{3} [(1.7 \times 10.2) + 4(3.9 + 7.89) + 2(4.2)] \\
 &= 22.5
 \end{aligned}$$

(b) (i)



(ii) At a point of inflexion the curve changes concavity. Since the sign of $f''(x)$ determines the concavity of a curve, we must have $f''(x) = 0$ at a point of inflexion AND the sign of $f''(x)$ changing as we move along the curve from one side of the point to the other.

10.

$$\begin{aligned}
 (c) \quad (i) \quad y &= x^3 - 6x^2 + 9x - 7 \quad \text{--- ①} \\
 \frac{dy}{dx} &= 3x^2 - 12x + 9
 \end{aligned}$$

Stationary point at $\frac{dy}{dx} = 0$

$$\begin{aligned}
 \text{i.e.} \quad 3x^2 - 12x + 9 &= 0 \\
 x^2 - 4x + 3 &= 0 \\
 (x-3)(x-1) &= 0 \\
 \therefore x &= 1, 3 \quad \text{and } \text{--- ②} \\
 y &= -3, -7
 \end{aligned}$$

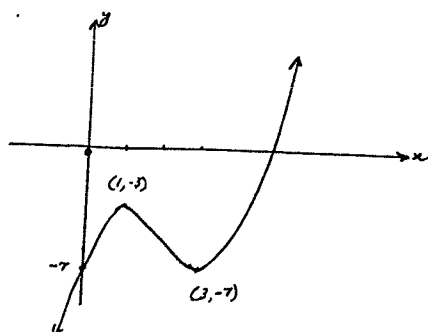
i.e. at $(1, -3)$ and $(3, -7)$

$$\frac{d^2y}{dx^2} = 6x - 12$$

at $(1, -3)$ $y'' = -6 \Rightarrow (1, -3)$ is a relative maximum turning point.

at $(3, -7)$ $y'' = 6 \Rightarrow (3, -7)$ is a relative minimum turning point.

(ii)

(iii) Increasing for $x < 1$ or $x > 3$

QUESTION 6:

$$(a) \quad R = R_0 e^{-kt} \quad R_0 = \text{initial amount}$$

$$(i) \quad \text{at } t = 5, R = \frac{1}{2} R_0$$

$$\therefore \frac{1}{2} R_0 = R_0 e^{-5k}$$

$$\Rightarrow e^{-5k} = \frac{1}{2}$$

$$-5k = \ln\left(\frac{1}{2}\right)$$

$$k = -\frac{1}{5} \ln\left(\frac{1}{2}\right)$$

$$= 0.138629 \dots$$

$$= 0.1386 \text{ (correct to 4 dec. places)}$$

$$(ii) \quad \text{when } R = \frac{3}{10} R_0$$

$$\frac{3}{10} R_0 = R_0 e^{-kt}$$

$$\frac{3}{10} = e^{-kt}$$

$$\Rightarrow -kt = \ln \frac{3}{10}$$

$$t = -\frac{1}{k} \ln\left(\frac{3}{10}\right)$$

$$= 8.684 \dots$$

$$= 8.7 \text{ (correct to 1 dec. place)}$$

$\therefore$ Another 3.7 years will elapse.

$$(b) (i) \text{ First } \$500 \text{ grows to } \$500 (1.005)^{36} = \$598.34$$

$$(ii) \text{ 1st } \$500 \text{ grows to } \$500 (1.005)^{36}$$

$$\text{2nd } \$500 \text{ grows to } \$500 (1.005)^{35}$$

$\vdots$

$$\text{last } \$500 \text{ grows to } \$500 (1.005)$$

$\therefore$ Lump sum value

$$= \$500 (1.005) + \$500 (1.005)^2 + \dots + \$500 (1.005)^{36}$$

$$\text{Geometric series } a = 500 (1.005) \quad r = 1.005 \quad n = 36$$

12.

$$= \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{500 (1.005) (1.005^{36} - 1)}{1.005 - 1} \text{ dollars} \quad \text{--- ①}$$

$$= \$100\,500 (1.005^{36} - 1)$$

$$(iii) \quad \$19\,766.39$$

$$(iv) \quad \text{① becomes}$$

$$22\,000 = \frac{R (1.005) (1.005^{36} - 1)}{1.005 - 1} \quad \text{where } \$R \text{ monthly saving.}$$

$$\text{i.e. } R = \frac{22\,000 (1.005 - 1)}{1.005 (1.005^{36} - 1)}$$

$$= \$556.50 \text{ (correct to nearest cent)}$$

QUESTION 7:

2) $1 + 5^x + 5^{2x} + \dots$
 Geometric series $a=1, r=5^x$
 $S = \frac{a}{1-r}$
 $\Rightarrow 5 = \frac{1}{1-5^x}$

in $5 - 5^{2x+1} = 1$
 $4 = 5^{2x+1}$

$x+1 = \log_5 4$

$x = \frac{\log_5 4}{\log_5 5} - 1$

$= -0.1386 \dots$

$= -0.139$ (correct to 3 dec. pl)

$x - y = 4 \quad \text{--- (1)}$

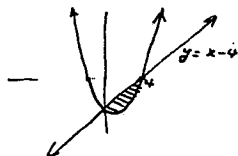
$y = x^2 - 3x - 4 \quad \text{--- (2)}$

(1) + (2): $x = 4 + x^2 - 3x - 4$

$0 = x^2 - 4x$

$0 = x(x-4)$

$x = 0, 4$



$A = \int_0^4 \{x-4 - (x^2-3x-4)\} dx$

$= \int_0^4 (4x - x^2) dx$

$= \left[2x^2 - \frac{1}{3}x^3 \right]_0^4$

$= \left(32 - \frac{64}{3} \right) - 0$

$= \frac{32}{3}$

$\therefore \Delta_{\text{area}} = \frac{32}{3} \text{ units}^2$

14.

(c) (i) $3x^2 + 2x + k$

$\Delta \equiv b^2 - 4ac$

$= 2^2 - 4(3)(k)$

$= 4 - 12k$

(ii) Real roots if $\Delta \geq 0$

in $4 - 12k \geq 0$

$4 \geq 12k$

$\therefore k \leq \frac{1}{3}$

(d) Let roots be $\alpha, 2\alpha$

$m\tilde{x} - px + 1 = 0$

$\therefore 3\tilde{x} = \frac{p}{m} \quad \text{--- (1)}$

$2\tilde{x} = \frac{1}{m} \quad \text{--- (2)}$

from (1) $\tilde{x} = \frac{p}{3m}$ sub in (2)

$\therefore 2 \times \left(\frac{p}{3m} \right)^2 = \frac{1}{m}$

$\frac{2p^2}{9m} = \frac{1}{m}$

$2p^2 = 9m$

QUESTION 8:

(a) (i) $v = \frac{dx}{dt} = 2t - 6$
 $\therefore x = t^2 - 6t + C$
 at $t=0, x=4$
 $\therefore 4 = C$
 $\therefore x = t^2 - 6t + 4$

(ii) comes to rest at $v=0$
 in $2t-6=0$
 $\therefore t=3$

at $t=3, x = 3^2 - 6(3) + 4$
 $= -5$

it comes to rest after 3 seconds and particle is 5m to the left of O.

(iii) at $t=0, x=4$
 $t=3, x=-5$
 $t=5, x=-1$ } $\therefore$ Total distance
 $= 9 + 4$
 $= 13 \text{ m}$

(iv) $\int_0^5 (2t-6) dt = \left[t^2 - 6t \right]_0^5$
 $= (25 - 30) - 0$
 $= -5$

$\int_0^5 (2t-6) dt$ is not the total distance travelled
 it is the change in displacement in the first 5 seconds. They would be the same if the particle did not stop and change directions in the first 5 seconds.

16.

(b) (i) $\frac{dG}{dt} = -\frac{3}{t+1}$

$\Rightarrow G = \int -\frac{3}{t+1} dt$

$G = -3 \ln(t+1) + C$

at $t=0, G=10$

$\therefore 10 = -3 \ln 1 + C$

$\therefore C = 10$

in $G = 10 - 3 \ln(t+1)$

(iii) Gas escapes $\Rightarrow G=0$

in $0 = 10 - 3 \ln(t+1)$

$3 \ln(t+1) = 10$

$\ln(t+1) = \frac{10}{3}$

$t+1 = e^{\frac{10}{3}}$

$t = e^{\frac{10}{3}} - 1$

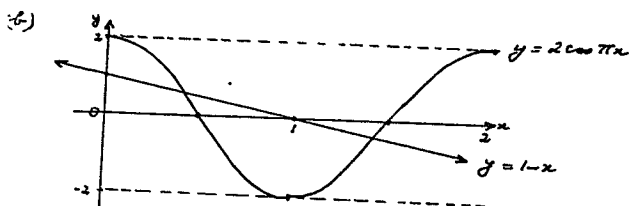
$= 27.03 \dots$

$\therefore$ Time is 27 seconds (correct to nearest second.)

QUESTION 9:

(a) (i) $\int \sin^2 x \cos x \, dx$
 $= \frac{1}{3} \sin^3 x + C$

(ii) $\int_0^{\frac{\pi}{2}} (2 \sin x - \cos 2x) \, dx$
 $= \left[-2 \cos x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$
 $= \left(-2 \cos \frac{\pi}{2} - \frac{1}{2} \sin \pi \right) - \left(-2 \cos 0 - \frac{1}{2} \sin 0 \right)$
 $= \left(-2 \cdot \frac{\sqrt{2}}{2} - \frac{1}{2} \cdot 0 \right) + 2$
 $= -\sqrt{2} + \frac{3}{2}$



There are 2 solutions for $0 \leq x \leq 2$.

(i) Perimeter $= (12 + 12 + 12 + 12 \cdot \frac{2\pi}{3}) \text{ cm}$
 $= (36 + 8\pi) \text{ cm}$

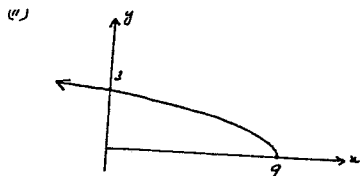
(ii) Area $= \text{area } \triangle POQ + \text{area sector } QRO$
 $= \left(\frac{1}{2} \cdot 12 \cdot 12 \sin \frac{\pi}{3} + \frac{1}{2} \cdot 12^2 \cdot \frac{2\pi}{3} \right) \text{ cm}^2$
 $= \left(72 \cdot \frac{\sqrt{3}}{2} + 48\pi \right) \text{ cm}^2$
 $= (36\sqrt{3} + 48\pi) \text{ cm}^2$

18.

QUESTION 10:

(a) (i) $y = \sqrt{9-x}$ Require $9-x \geq 0$
 $x \leq 9$

$\therefore$ Domain is $\{x: x \leq 9\}$
 Range is $\{y: y \geq 0\}$



(iii) $V = \pi \int_0^3 x^2 \, dy$

$y = \sqrt{9-x}$
 $y^2 = 9-x$
 $x = 9-y^2$
 $x^2 = 81 - 18y^2 + y^4$

$= \pi \int_0^3 (81 - 18y^2 + y^4) \, dy$
 $= \pi \left[81y - 6y^3 + \frac{y^5}{5} \right]_0^3$
 $= \pi \left[243 - 162 + \frac{243}{5} - 0 \right]$
 $= \frac{648\pi}{5}$

$\therefore$ Volume is $\frac{648\pi}{5} \text{ units}^3$

19.

(b) (i) $A = \frac{1}{2} \text{ base} \times \perp \text{ height}$
 $= \frac{1}{2} \cdot t \cdot \sqrt{9-t^2}$
 $= \frac{t}{2} (9-t^2)^{\frac{1}{2}}$

(ii) $\frac{dA}{dt} = (9-t^2)^{\frac{1}{2}} \cdot \frac{1}{2} + \frac{t}{2} \cdot \frac{1}{2} (9-t^2)^{-\frac{1}{2}} \cdot (-2t)$
 $= \frac{\sqrt{9-t^2}}{2} - \frac{t^2}{2\sqrt{9-t^2}}$

stationary point occurs when $\frac{dA}{dt} = 0$

ie $\frac{\sqrt{9-t^2}}{2} - \frac{t^2}{2\sqrt{9-t^2}} = 0$

ie $\frac{\sqrt{9-t^2}}{2} = \frac{t^2}{2\sqrt{9-t^2}}$

$\Rightarrow \frac{2(9-t^2)}{18-2t^2} = \frac{2t^2}{18-2t^2}$
 $18-2t^2 = 2t^2$
 $18 = 4t^2$
 $t^2 = \frac{18}{4}$

$\therefore t = \frac{3\sqrt{2}}{2}$ (since $t \geq 0$)

TEST:

t	2.1	$\frac{3\sqrt{2}}{2}$	2.2
$\frac{dA}{dt}$	0.6	0	-0.2

NOTE: must give values for this test!!

$\therefore$ maximum area occurs at $t = \frac{3\sqrt{2}}{2}$

$\therefore$ Co-ordinates of P are $\left(\frac{3\sqrt{2}}{2}, \sqrt{9-\frac{9}{2}} \right)$
 $= \left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2} \right)$